Cylinder liner bore survey

Unit 3 · MV Example (simulated vessel) · 2-stroke crosshead, 980 bore (simulated)

Vessel / IMO	MV Example (simulated vessel) / —
Engine	2-stroke crosshead, 980 bore (simulated)
Engine serial	—
Cylinder unit	3
Running hours / since overhaul	38 450 (example) / 11 200 (example)
Client	— (example)
Location / date	Dubai anchorage (simulated) / 2026-10-05
Surveyor	UMA Marine — name to be entered

Instrument	LinerScan LS-0001, fw 0.1.0-sim
Optical module	preset L · calibration CAL-SIM-0001 (2026-10-01)
Nominal bore	Ø 980.00 mm
Liner temperature	62 °C (jacket water) · instrument 60.6 °C
Scan	1 stroke BDC → 15.3° BTDC in 261 s · 2.26 M points
Scanned depth	0 – 2664 mm below liner top
Device placement	offset 16.8 mm, tilt 0.33° / -0.22° (removed by fit)
Orientation	FORE at 92.5° device angle (scavenge-port pattern)

Key results

LARGEST DIAMETER
982.41
mm at 195 mm depth, 20°/200°

INCREASE VS NOMINAL
+2.41
mm (as measured) = 31% of limit*

MAX OVALITY |PS−FA|
0.19
mm at 165 mm · 37% of limit*

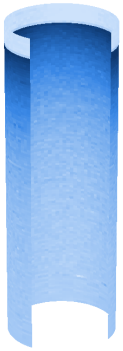
MAX D MAX − D MIN
0.35
mm at 195 mm depth

Condition against limits (draft)

Criterion	Measured (worst band)	Limit*	Used	Assessment
Diameter increase (D max − nominal)	2.41 mm @ 195 mm	7.84 mm	31%	WITHIN LIMIT
Ovality |D PS − D FA|	0.19 mm @ 165 mm	0.50 mm	37%	WITHIN LIMIT

* Limits are **placeholders** (0.8% of bore for diameter increase, 0.50 mm ovality) and must be replaced with the engine maker's values before this report is issued. The wording and the 80% "approaching" threshold are a draft for UMA's decision.
Diameters are stated **as measured** at a liner temperature of 62 °C; no correction to 20 °C is applied. For reference, thermal expansion of a cast-iron liner from 20 °C to 62 °C is about +0.45 mm on the diameter and is contained in the increase shown above.
Expanded uncertainty of a band diameter U(k=2) ≤ 0.020 mm (statistical + systematic budget, see page 4).

3D scan (cut-away), colour as wall map



Scan at a glance

One continuous stroke recorded 2.26 million wall points from 0 to 2664 mm below the liner top: 89 bands of 30 mm, each measured all around the bore.

- Coverage: at least 100% of 1° sectors in every non-port band.
- Outliers marked: 0.20% of points.
- Ring check: before job +2 µm; after job -3 µm
- Flagged bands: 11 (ports, ridge, shape) — see appendix.
- Crush safety: no warning or alarm.
- Thermal compensation active, instrument at 60.6 °C.

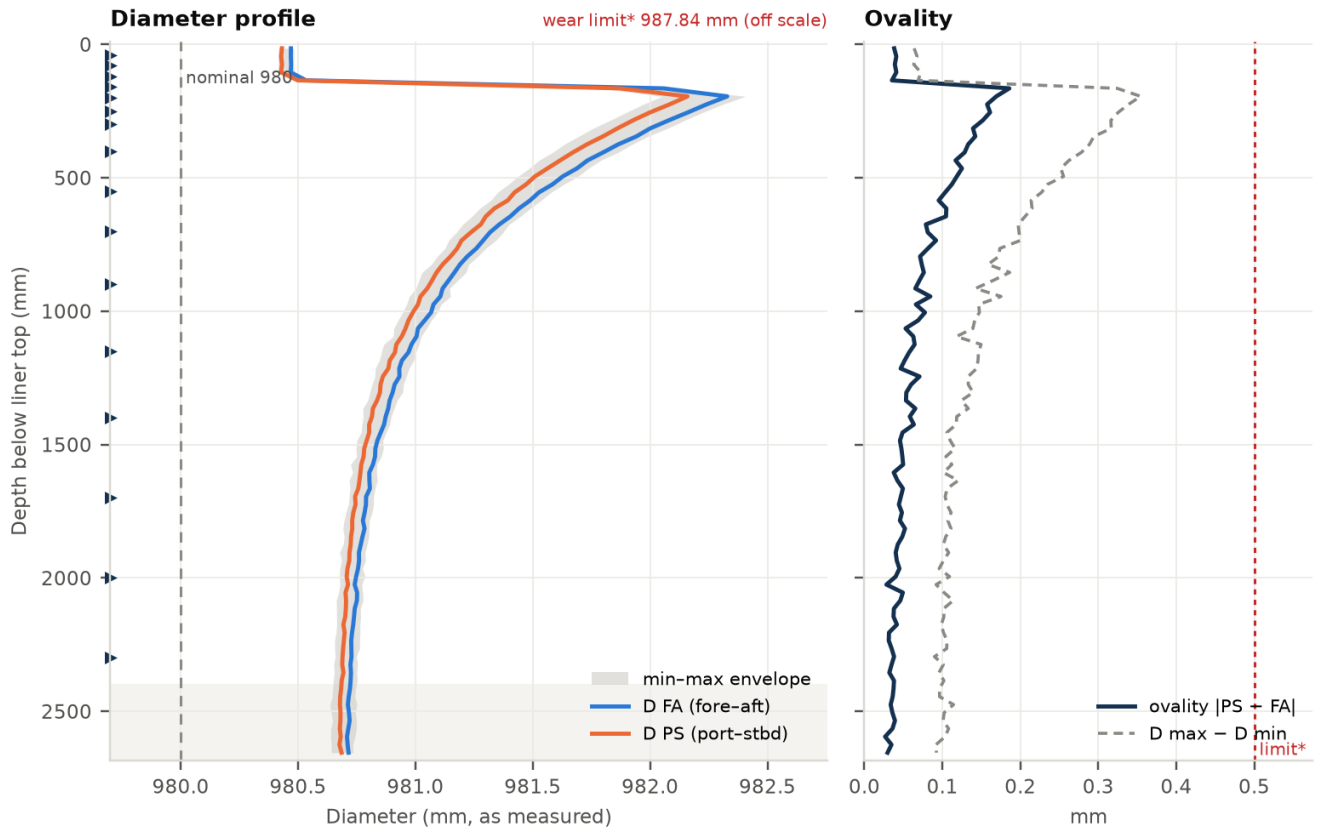
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Diameters at the standard measuring positions

Each value is a 30 mm band (±15 mm about the stated depth) fitted with 3D data all around the bore; diameters are taken across the fitted liner centre. Angles from FORE, counter-clockwise seen from above (PORT 90°).

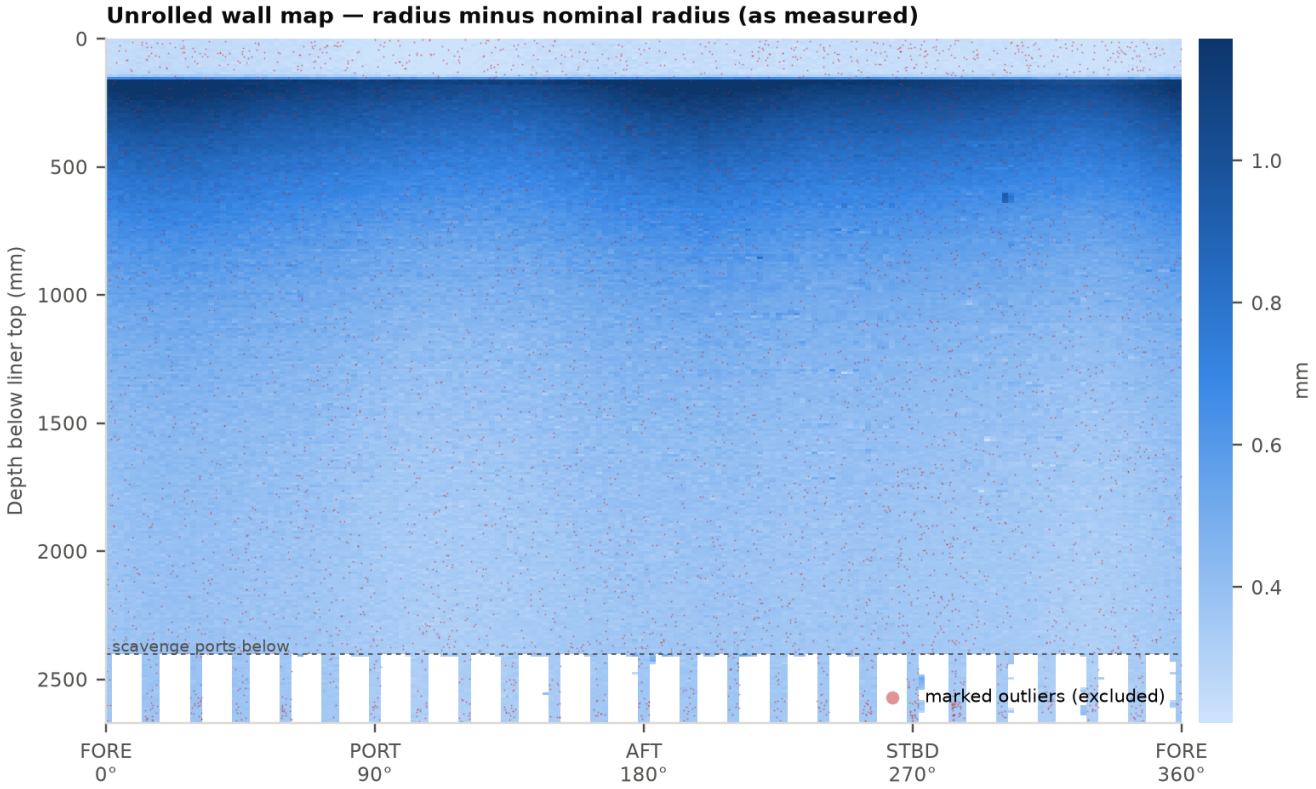
Pos	Depth mm	D FA mm	D PS mm	Ovality |PS−FA|	D max mm	at °	D min mm	at °	max−min mm	Δ nominal mm	U(k=2) mm	Note
1	40	980.47	980.43	0.04	980.48	25	980.42	103	0.07	+0.45	0.017	
2	80	980.47	980.43	0.04	980.48	16	980.42	103	0.06	+0.45	0.017	
3	120	980.47	980.44	0.03	980.49	24	980.42	103	0.07	+0.46	0.017	
4	160	981.78	981.63	0.16	981.83	17	981.55	137	0.29	+1.68	0.019	shape, steep
5	200	982.31	982.15	0.17	982.39	19	982.04	137	0.36	+2.20	0.018	
6	250	982.17	982.01	0.16	982.23	20	981.91	140	0.33	+2.06	0.018	
7	300	982.03	981.89	0.14	982.11	20	981.79	139	0.32	+1.93	0.018	
8	400	981.81	981.68	0.13	981.87	18	981.59	136	0.28	+1.72	0.018	
9	550	981.54	981.43	0.10	981.59	20	981.36	139	0.23	+1.46	0.018	
10	700	981.33	981.24	0.08	981.37	21	981.18	140	0.20	+1.27	0.019	
11	900	981.14	981.06	0.07	981.16	21	981.00	138	0.17	+1.08	0.019	
12	1150	980.97	980.91	0.06	981.01	27	980.85	143	0.16	+0.93	0.019	
13	1400	980.87	980.81	0.06	980.89	17	980.78	138	0.11	+0.83	0.018	
14	1700	980.80	980.74	0.05	980.82	20	980.72	143	0.10	+0.77	0.018	
15	2000	980.75	980.71	0.04	980.79	28	980.67	142	0.11	+0.73	0.018	
16	2300	980.73	980.69	0.04	980.75	18	980.66	140	0.09	+0.70	0.017	

Notes: ports = band in the scavenge-port zone (diameters across the port bars only); steep = diameter changes > 0.1 mm across the band (ring-reversal ridge), the value depends on the exact height; shape = cross-section not elliptical (ellipse cross-check differs from the Fourier result by > 0.05 mm). Δ nominal = mean diameter − nominal.

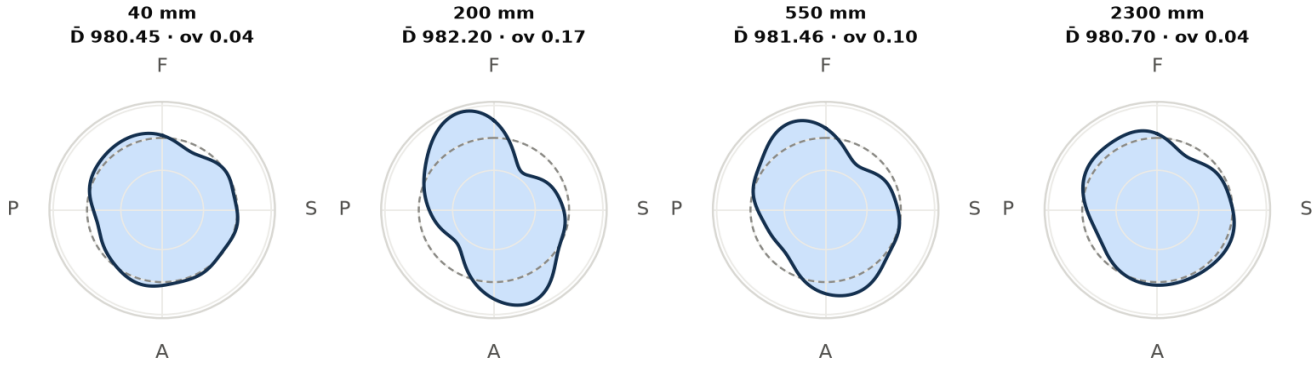


Arrows mark the standard positions; shaded = scavenge-port zone. Red dashed lines are placeholder limits (*).

Wall map and cross-sections



Every scan point within each 30 mm band, binned 2° × 10 mm. Red dots are points marked as outliers (beyond max(5·1.4826·MAD, 0.1 mm) of the band fit): 4,471 of 2,263,395 (0.20%). They are excluded from the fits but kept in the data set; clusters of them mark real local features such as scuffing.



Shape about the mean circle (dashed). Grid rings at ±0.11 mm radial — deviations magnified for display.

Quality and traceability

Triangulation model	hyperbolic, surface (leave-one-out RMS 15.6 µm; chosen by the ≥20 % and ≥1 µm rule)
Thermal compensation	on · reference 21 °C · camera 60.6 °C during scan
Head geometry	R0 32.277 mm, h 4.30 mm (ring Ø870, fit rms 71 µm)
Ring checks	before job: +1.8 µm at 34 °C; after job: -3.2 µm at 41 °C
Height reference	laser rangefinder (scale 0.9996), anchored at BDC and at the stop: liner-top edge seen at the stop (1549 frames; flywheel reading differs by +2.9 mm)
Orientation	scavenge-port pattern (strength 0.23), placement 90.0°
Coverage (non-port bands)	min 100% of 1° sectors, largest gap 0°
Outliers	4,471 marked and excluded (0.20%)
Flagged bands	11 of 89 (see appendix)
Crush safety	no warnings · BLE status 964/1227 packets

Uncertainty contribution (diameter)	u (mm)
Point noise averaged over a band (Fourier covariance, per band)	≤ 0.0063
Systematic: calibration, thermal model, encoder, height (simulation-validated)	0.0080
Expanded U (k = 2), worst band	0.020

Per-position fit quality

Depth mm	Points	Outliers	Cover %	Max gap °	Tilt x/y °	Centre offset mm	σ_r mm	Ellipse Δ mm	Flags
40	78,667	142	100	0	0.34 / -0.21	16.8	0.100	0.007	
80	55,942	115	100	0	0.35 / -0.21	16.8	0.100	0.010	
120	38,375	89	100	0	0.35 / -0.21	16.8	0.100	0.005	
160	34,615	66	100	0	0.33 / -0.22	16.8	0.164	0.102	non-elliptical, steep
200	30,811	62	100	0	0.35 / -0.21	16.8	0.102	0.048	
250	27,467	51	100	0	0.35 / -0.22	16.8	0.101	0.037	
300	25,492	49	100	0	0.35 / -0.22	16.8	0.102	0.042	
400	22,650	51	100	0	0.35 / -0.23	16.8	0.102	0.037	
550	21,515	42	100	0	0.34 / -0.22	16.8	0.100	0.020	
700	19,467	46	100	0	0.33 / -0.23	16.8	0.103	0.024	
900	18,571	43	100	0	0.33 / -0.22	16.7	0.103	0.021	
1150	17,459	33	100	0	0.34 / -0.22	16.7	0.102	0.013	
1400	20,355	35	100	0	0.36 / -0.21	16.7	0.100	0.002	
1700	21,774	44	100	0	0.33 / -0.21	16.7	0.100	0.003	
2000	26,459	49	100	0	0.33 / -0.23	16.7	0.101	0.011	
2300	36,284	67	100	0	0.35 / -0.23	16.6	0.101	0.010	

Scan timeline

Time s	Event
0.0	SOAK
900.0	ARMED after 900 s soak (dT/dt 0.39 K/min)
938.7	SCAN piston rising
1199.9	STOPPED clearance 53mm
1204.0	DONE 60800 frames

Ellipse cross-check: a direct least-squares ellipse is fitted to every band; where its major and minor diameters differ from the Fourier D max / D min by more than 0.05 mm the band is flagged as non-elliptical (lobed or locally worn cross-section).

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Method and definitions

Principle. A 405 nm laser line is projected on the liner wall and observed by a camera at a fixed baseline. In every image row the sub-pixel position of the line gives the distance to the wall (laser-line triangulation, z = a/(u − u0) + b per row). The head sweeps ±190° while the turning gear raises the piston from BDC, so one stroke covers the whole wall.
Geometry. Points are placed with the calibrated arm offset R0, beam offset h and the ray elevation; each 30 mm band is fitted with a local cylinder (removes device tilt and off-centre placement), a circle about the band axis and an order-8 Fourier radius profile. D(α) = ρ(α) + ρ(α+180°) across the fitted centre — never through the device axis.
Definitions. D FA = diameter fore–aft, D PS = diameter port–starboard, ovality = |D PS − D FA| (UMA definition). D max / D min are the largest / smallest diameter over all directions, with their angles; max − min is reported separately. Depths are below the liner top.
Height. Crown position from the upward laser distance module, scaled between BDC and the stop position (liner-top edge seen by the scanner when it is in view, cross-checked with the flywheel reading).
Safety. Planned stop crank angle (primary) and two upward multizone ToF sensors with buzzer (secondary). Safety does not depend on the radio link.

Remarks

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Sign-off

Surveyor (UMA Marine)	Chief Engineer	Superintendent / Class
Name / signature / date	Name / signature / date	Name / signature / date

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Appendix A — every 30 mm band

Depth	D FA	D PS	Ovality	D max	°	D min	°	max−min	Points	Outl.	Cover %	U(k=2)	Flags
15	980.47	980.43	0.04	980.48	26	980.42	105	0.06	63,598	111	100	0.017	
45	980.47	980.43	0.04	980.48	25	980.42	102	0.07	81,831	144	100	0.017	
75	980.47	980.43	0.04	980.48	21	980.42	104	0.06	65,178	132	100	0.017	
105	980.47	980.43	0.04	980.49	25	980.41	103	0.07	39,873	84	100	0.017	
135	980.53	980.50	0.04	980.55	24	980.48	105	0.07	36,377	77	100	0.017	steep
165	982.06	981.87	0.19	982.11	17	981.79	139	0.32	34,081	60	100	0.020	non-elliptical, steep
195	982.33	982.16	0.17	982.41	20	982.05	138	0.35	31,228	60	100	0.018	
225	982.24	982.08	0.16	982.31	18	981.97	137	0.34	29,810	53	100	0.018	
255	982.16	982.00	0.16	982.23	19	981.90	139	0.33	27,446	49	100	0.018	
285	982.08	981.93	0.15	982.15	18	981.83	139	0.32	26,215	62	100	0.018	
315	982.00	981.86	0.14	982.07	20	981.76	140	0.32	24,219	43	100	0.018	
345	981.94	981.80	0.14	981.99	18	981.70	140	0.29	24,311	48	100	0.018	
375	981.86	981.73	0.13	981.92	20	981.63	138	0.29	22,769	49	100	0.018	
405	981.80	981.67	0.13	981.86	18	981.59	137	0.28	22,105	45	100	0.018	
435	981.73	981.62	0.12	981.79	20	981.53	139	0.26	22,872	41	100	0.018	
465	981.69	981.56	0.13	981.74	19	981.49	139	0.25	21,358	58	100	0.018	
495	981.63	981.51	0.12	981.69	20	981.44	138	0.26	21,727	39	100	0.018	
525	981.58	981.47	0.11	981.63	19	981.40	142	0.23	21,999	37	100	0.018	
555	981.53	981.42	0.10	981.58	19	981.35	138	0.23	20,947	40	100	0.018	
585	981.49	981.39	0.10	981.54	19	981.32	137	0.21	21,692	40	100	0.018	
615	981.44	981.34	0.11	981.49	24	981.27	143	0.22	20,705	44	100	0.018	
645	981.40	981.30	0.10	981.45	17	981.24	142	0.21	21,240	35	100	0.018	
675	981.36	981.28	0.08	981.40	21	981.20	140	0.20	21,040	44	100	0.018	
705	981.32	981.24	0.08	981.37	21	981.18	139	0.20	19,217	48	100	0.018	
735	981.29	981.20	0.09	981.34	20	981.14	136	0.20	21,434	45	100	0.018	
765	981.26	981.18	0.08	981.30	17	981.13	137	0.17	19,117	48	100	0.018	
795	981.22	981.15	0.07	981.27	21	981.09	130	0.17	19,232	41	100	0.019	
825	981.19	981.12	0.07	981.23	20	981.07	142	0.16	19,374	38	100	0.018	
855	981.17	981.09	0.08	981.22	23	981.03	138	0.19	18,585	34	100	0.019	
885	981.14	981.07	0.07	981.18	23	981.01	138	0.17	18,773	38	100	0.019	
915	981.12	981.05	0.07	981.15	22	981.00	141	0.14	17,320	45	100	0.019	
945	981.11	981.02	0.08	981.15	23	980.97	143	0.18	19,377	33	100	0.019	
975	981.08	981.01	0.07	981.11	25	980.96	144	0.15	17,242	36	100	0.019	
1005	981.07	980.99	0.08	981.09	22	980.95	142	0.15	18,315	30	100	0.019	
1035	981.04	980.97	0.07	981.07	21	980.93	141	0.14	16,826	29	100	0.019	
1065	981.01	980.96	0.05	981.05	24	980.91	142	0.14	19,217	33	100	0.019	
1095	981.00	980.94	0.06	981.02	14	980.91	147	0.12	17,217	38	100	0.019	
1125	980.98	980.92	0.06	981.01	21	980.87	142	0.15	18,611	43	100	0.019	
1155	980.97	980.91	0.06	981.00	27	980.86	141	0.15	17,926	32	100	0.019	
1185	980.94	980.89	0.05	980.99	24	980.85	139	0.15	19,465	35	100	0.018	
1215	980.93	980.89	0.05	980.97	27	980.83	141	0.14	18,598	39	100	0.018	
1245	980.93	980.86	0.07	980.96	19	980.83	145	0.13	18,516	28	100	0.019	
1275	980.91	980.85	0.06	980.95	20	980.82	140	0.13	20,041	51	100	0.018	
1305	980.90	980.85	0.05	980.95	25	980.81	141	0.14	18,844	47	100	0.018	
1335	980.89	980.84	0.05	980.92	27	980.80	143	0.12	20,275	38	100	0.019	
1365	980.88	980.82	0.07	980.91	22	980.78	143	0.13	18,891	42	100	0.018	
1395	980.87	980.82	0.06	980.90	18	980.78	139	0.12	20,963	42	100	0.018	
1425	980.87	980.80	0.06	980.89	23	980.77	146	0.12	19,479	37	100	0.018	
1455	980.85	980.80	0.05	980.88	25	980.77	140	0.11	19,121	35	100	0.019	
1485	980.84	980.79	0.05	980.86	24	980.75	145	0.11	20,667	43	100	0.018	
1515	980.83	980.78	0.05	980.86	24	980.75	144	0.12	18,942	27	100	0.018	
1545	980.83	980.78	0.05	980.85	21	980.75	142	0.10	20,241	38	100	0.019	
1575	980.82	980.77	0.05	980.84	23	980.72	142	0.11	18,938	31	100	0.018	
1605	980.80	980.77	0.04	980.83	28	980.73	138	0.10	20,474	44	100	0.018	
1635	980.80	980.76	0.04	980.84	21	980.72	134	0.12	19,678	44	100	0.018	
1665	980.81	980.76	0.05	980.83	25	980.72	138	0.11	19,201	45	100	0.019	

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Depth	D FA	D PS	Ovality	D max	°	D min	°	max−min	Points	Outl.	Cover %	U(k=2)	Flags
1695	980.79	980.74	0.05	980.82	23	980.72	140	0.10	21,561	45	100	0.018	
1725	980.79	980.75	0.04	980.82	24	980.72	140	0.11	20,068	42	100	0.018	
1755	980.78	980.74	0.05	980.82	21	980.70	144	0.11	21,179	39	100	0.018	
1785	980.78	980.73	0.05	980.80	23	980.70	139	0.11	22,065	50	100	0.018	
1815	980.78	980.73	0.05	980.80	28	980.69	141	0.11	22,112	30	100	0.018	
1845	980.77	980.73	0.05	980.80	28	980.70	141	0.10	23,183	54	100	0.018	
1875	980.77	980.72	0.04	980.79	23	980.69	142	0.10	24,024	40	100	0.018	
1905	980.76	980.72	0.04	980.79	24	980.68	144	0.11	24,502	55	100	0.018	
1935	980.76	980.72	0.04	980.79	20	980.69	141	0.10	24,775	46	100	0.018	
1965	980.76	980.71	0.04	980.77	26	980.68	145	0.10	26,245	51	100	0.018	
1995	980.75	980.71	0.04	980.79	27	980.68	141	0.11	26,463	51	100	0.018	
2025	980.74	980.71	0.03	980.77	29	980.68	110	0.09	26,310	50	100	0.018	
2055	980.75	980.70	0.05	980.78	20	980.67	143	0.10	27,637	50	100	0.018	
2085	980.75	980.70	0.05	980.78	21	980.66	142	0.11	26,833	47	100	0.018	
2115	980.74	980.70	0.04	980.77	29	980.66	140	0.10	27,952	59	100	0.018	
2145	980.74	980.70	0.04	980.77	24	980.66	142	0.10	28,871	42	100	0.018	
2175	980.74	980.69	0.04	980.76	28	980.67	143	0.10	29,391	60	100	0.018	
2205	980.73	980.70	0.03	980.76	24	980.66	142	0.10	31,245	44	100	0.018	
2235	980.73	980.70	0.03	980.76	28	980.66	143	0.11	31,806	64	100	0.018	
2265	980.73	980.69	0.04	980.76	26	980.66	144	0.11	34,152	64	100	0.017	
2295	980.73	980.69	0.04	980.75	19	980.66	116	0.09	35,933	64	100	0.017	
2325	980.72	980.69	0.03	980.75	27	980.65	142	0.10	39,369	65	100	0.017	
2355	980.73	980.69	0.03	980.75	22	980.65	143	0.10	41,743	99	100	0.017	
2385	980.72	980.69	0.04	980.75	26	980.65	145	0.10	44,734	99	100	0.017	
2415	980.72	980.68	0.04	980.74	28	980.65	142	0.10	12,896	26	41	0.020	ports
2445	980.72	980.68	0.04	980.74	26	980.65	141	0.10	13,379	35	28	0.019	ports
2475	980.71	980.68	0.03	980.75	24	980.64	145	0.11	14,619	22	28	0.019	ports
2505	980.72	980.68	0.04	980.74	32	980.64	142	0.10	16,415	41	28	0.019	ports
2535	980.72	980.68	0.04	980.75	19	980.65	148	0.10	19,908	43	28	0.018	ports
2565	980.71	980.68	0.04	980.75	27	980.64	143	0.11	26,281	43	28	0.018	ports
2595	980.71	980.68	0.03	980.74	30	980.64	144	0.10	48,817	105	29	0.017	ports
2625	980.71	980.68	0.03	980.73	26	980.64	144	0.09	46,334	105	28	0.017	ports
2655	980.72	980.69	0.03	980.73	24	980.64	145	0.09	28,975	69	28	0.018	ports

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